RAJALAKSHMI ENGINEERING COLLEGE

Harshini M 251801097-2025-AIDS

H2

Started on	Saturday, 12 September 2026, 2:26 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:28 PM
Time taken	2 mins 37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

#define MAX(a, b) ((a) > (b) ? (a) : (b))

int longestCommonSubsequence(const char *s1, const char *s2) {
    int m = strlen(s1);
    int n = strlen(s2);

    int dp[m + 1][n + 1];

    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0;
            }
            else if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            }
            else {
                dp[i][j] = MAX(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }
    return dp[m][n];
}

int main() {
    // Arrays to hold the input strings (adjust size if strings can be longer)
    char s1[1000];
    char s2[1000];

    // Read the two strings from standard input
    if (scanf("%s %s", s1, s2) == 2) {
        // Output ONLY the final integer number as expected by the grader
        printf("%d\n", longestCommonSubsequence(s1, s2));
    }

    return 0;
}

```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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